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ICH 252 — Past Questions & Materials
by Renance

Course: ICH 252
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — ICH 252 - Process Science II

Level 200 • Second Semester • PastQuestion

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**DEPARTMENT OF CHEMISTRY
COLLEGE OF SCIENCE
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
SECOND SEMESTER EXAMINATION FOR 2024/2025 ACADEMIC SESSION
COURSE TITLE: PROCESS SCIENCE II
COURSE CODE: ICH 252**

**UNITS: 3
TIME: 2 HOURS**

INSTRUCTION: ANSWER ANY THREE (3) QUESTIONS

QUESTION ONE

(a) Describe the term Mass Transfer processes **7 marks**

(b) Explain the mechanism behind mass transfer. **8 marks**

(c) Ammonia gas is diffusing at a constant rate through a layer of stagnant air 1mm thick. Conditions are such that the gas contains 50% by volume of ammonia at one boundary of the stagnant layer. The ammonia diffusing to the other boundary is quickly absorbed and the concentration is negligible at that plane. The temperature is 295k and also the atmospheric pressure, and under these conditions, the diffusivity of ammonia in air is $0.18\text{cm}^2/\text{s}$. Estimate the rate of diffusion of ammonia through the layer. **10 marks**

QUESTION TWO

(a) State the Fick's law **5 marks**

(b) In an air-carbon dioxide mixture at 298k and 202.6kpa, the concentration of CO_2 at two planes (2mm) apart are 25vol% and 15vol% respectively. The diffusivity of CO_2 in air at 298k and 202.6kpa is $8.2 \times 10^{-6} \text{m}^2/\text{s}$. calculate the rate of transfer of CO_2 across the two planes, assuming:

(i) Equimolecular counter diffusion.

(ii) Diffusion of CO_2 through a stagnant air layer. **10 marks**

(c) Determine the diffusivity of (i) CO_2 (ii) O_2 (iii) N_2
 CO_2 : 28.5%, O_2 : 15%, N_2 : 56.5%

The gas mixture is at 273k and $1.2 \times 10^5 \text{Pa}$. The binary diffusivity values are given as (at 273k).

$D_{12} = 1.874 \text{m}^2/\text{Pa}\cdot\text{s}$

$D_{13} = 1.945 \text{m}^2/\text{Pa}\cdot\text{s}$

$D_{23} = 1.834 \text{m}^2/\text{Pa}\cdot\text{s}$

10 marks

QUESTION THREE

(a) With the aid of a diagram discuss filtration and sedimentation. **5 marks**

(b) Explain stoichiometry. Differentiate between stoichiometric coefficients and Stoichiometric ratios. **10 marks**

(c) What is rate determining step? Explain four rate determining steps under gas film and liquid film **10 marks**

QUESTION FOUR

- (a) Explain the principle of solvent extraction in mixer-settlers (5 marks)
- (b) Discuss the main stages involved and state one advantage and one limitation of using mixer-settlers (12 marks)
- (c) State two major industrial applications of mixer-settlers and two of extraction columns. (8 marks)

QUESTION FIVE

- (a) A distillation column separates a binary mixture. The feed is at its boiling point (saturated liquid), with mole fraction of component A=0.5. The desired distillate composition is 0.95 and the bottoms composition is 0.05. Assuming total reflux, estimate the minimum number of ideal stages using the Fenske equation ($\alpha = 2.4$) (9 marks).
- (b) Give two methods used to determine the number of ideal stages in distillation and explain the relevance of Kremser equation? (7 marks).
- (c) Two stage counter current (ideal stages) in which aqueous feed flow is 1200L, initial solute concentration in feed is 15.0 g/L, per stage solvent flow $S=400\text{L}$, and the distribution coefficient is 1.8. Find the final raffinate X_2 and extraction fraction (9 marks).

THE END